Salifort Motors

Employee Retention Project

Issue/ Problem

Salifort Motors seeks to improve employee retention and answer the following question:

What's likely to make the employee leave the company?

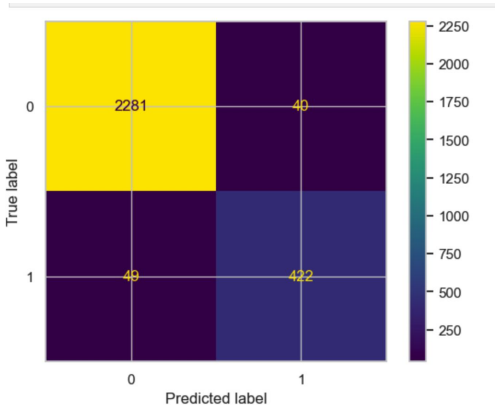
Response

Since the variable we are seeking to predict is categorical, the team could build either a logistic regression or a tree-based machine learning model.

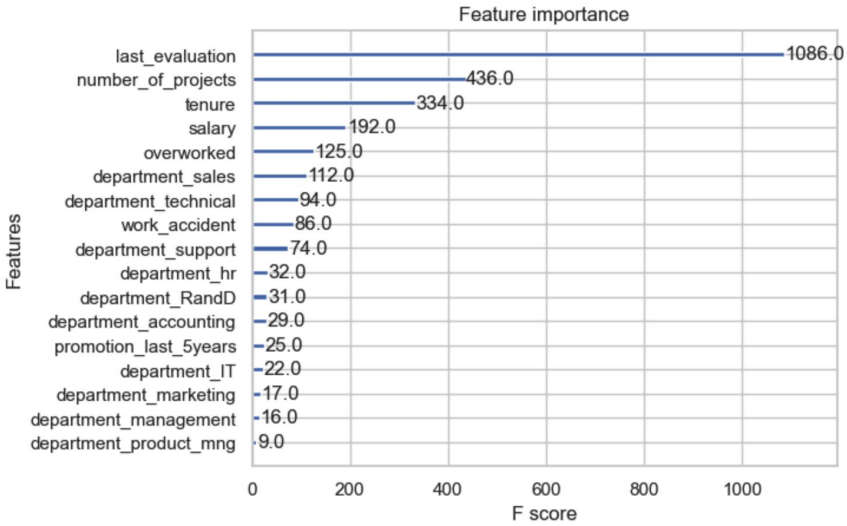
The XGBoost model outperform the Logistic Regression model as a result.

Impact

This model helps predict whether an employee will leave and identify which factors are most influential. These insights can help HR make decisions to improve employee retention.



F1 score final XGB model: 0.9046087888531619
Recall score final XGB model: 0.8959660297239915
Precision score final XGB model: 0.9134199134199135
Accuracy score final XGB model: 0.9681232091690545



In the Final XGBoost model above, `***last_evaluation***`, `***number_project***`, `***tenure***`, `***overworked***`, ***and*** `***salary***` have the highest importance. These variables are most helpful in predicting the outcome variable, `*left*`.

Insights / Next steps

- Cap the number of projects that employees can work on.
- Promote employees who have been showing good evaluation scores and work at the company for at least 4 years.
- Reduce working hours or compensate better for overtime.
- Evaluation scores shouldn't solely rely on working hours.
- If employees aren't familiar with the company's overtime pay policies, inform them about this. If the expectations around workload and time off aren't explicit, make them clear.
- High evaluation scores should not be reserved for employees who work 200+ hours per month. Consider a proportionate scale for rewarding employees who contribute more/put in more effort.